Marmara University
Engineering Faculty, Electrical and Electronics Engineering
EE4077 Fundamentals of Machine Learning, Final Exam Solutions

1) (a) The sample mean is $\frac{1}{6}\sum_{i=1}^{6} x_i = \frac{1}{6}\begin{bmatrix}9\\9\end{bmatrix} = \begin{bmatrix}1.5\\1.5\end{bmatrix}$ (2)

By subtracting the mean from each sample we form the centered design matrix $\Rightarrow X = \begin{bmatrix}-1.5 & -1.5\\ -1.5 & 0.5\\ 0.5 & -1.5\\ -0.5 & 1.5\\ 1.5 & -0.5\\ 1.5 & 1.5\end{bmatrix}$ (2)

(b) Principal components are the eigenvectors of the matrix $X^TX = \begin{bmatrix}9.5 & 1.5\\ 1.5 & 9.5\end{bmatrix}$

The charcteristic polynomial is $\det(\lambda I - X^TX) = \det\begin{bmatrix}\lambda-9.5 & 1.5\\ 1.5 & \lambda-9.5\end{bmatrix}$ (2)

$= (\lambda-9.5)^2 - 1.5^2$ (2)

Hence the eigenvalues of X^TX are $\lambda_1 = 8$, $\lambda_2 = 11$.

$X^TXv_1 = \lambda_1 v_1 = 8v_1 \Rightarrow v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$ (3)

$X^TXv_2 = \lambda_2 v_2 = 11v_2 \Rightarrow v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$ (3)

(c) PCA picks the principal component v_2 with the largest eigenvalue (2)

(d) We prefer v_2 because it maximizes the variance of the sample points after they are projected onto a line parallel to v_2. (2)

(e) It minimizes the sum of the squared distances from each sample point to the projection on that line (2)

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2. (a) $w_1 = w_2 = 5$ will work (8)

(b) $w_1 = w_2 = 20$ will work (8)

(c) NOR problem is not linearly separable. A linear decision surface can not be used for a non-linear optimization problem (4)

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3 (a) $9 \times 6 = 54$ parameters (5)

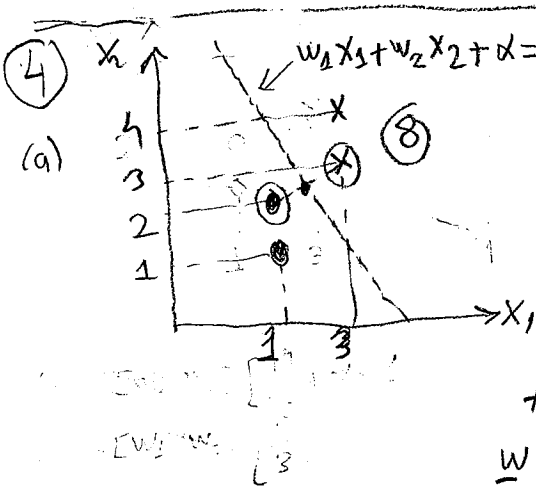
(b) $16 \times 15 \times 15 = 1350$ ReLU operations (5)

(c) $\underbrace{9\times6}_{\text{CNN}} + \underbrace{6\times15\times15\times10}_{\text{FCN}} = 13554$ parameters (5)

(d) FCN has too many weights to learn (5)

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4 (a)



(8)

(b) $[w_1\ w_2]\begin{bmatrix}1\\2\end{bmatrix} + \alpha = 1$ (3)

$[w_1\ w_2]\begin{bmatrix}3\\3\end{bmatrix} + \alpha = -1$ (3)

$[w_1\ w_2]\begin{bmatrix}2\\2.5\end{bmatrix} + \alpha = 0$ (3)

Solving the above three equations we get

$\underline{w} = \begin{bmatrix}w_1\\w_2\end{bmatrix} = \begin{bmatrix}-0.8\\-0.4\end{bmatrix}$ $\alpha = 2.6$ (3)

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5. (a) False. Image captioning and language translation are sequence-to-sequence conversion for which RNN must be used. (5)

(b) False. Backpropagation through time algorithm is used. (5)

(c) True. k-means does not require true labels (5)

(d) False PCA is used for dimensionality reduction. (5)